

UNIT V | Plant Reproduction, Growth and Propagation (12 Lectures)

1.1 Plant Propagation — Types of Propagules

Propagule	Type	Example
Rhizome	Underground stem	Ginger, Turmeric, Lotus
Tuber	Underground stem (swollen tip)	Potato
Bulb	Modified stem with scale leaves	Onion, Garlic
Corm	Solid underground stem	Colocasia, Saffron
Runner	Sub-aerial stem on soil	Grass, Strawberry, Oxalis
Sucker	Underground stem (oblique)	Banana, Mint, Chrysanthemum
Bulbil	Fleshy bud	Agave, Garlic
Foliar buds	Leaf margin buds	Bryophyllum
Cutting (artificial)	Stem/leaf/root piece	Rose, Hibiscus, Begonia
Grafting (artificial)	Stock + Scion union	Mango, Apple, Rose

1.2 Sexual and Asexual Reproduction

Double Fertilization (unique to Angiosperms — discovered by Nawaschin, 1898):

Sperm 1 (n) + Egg (n) → Zygote (2n) → Embryo
Sperm 2 (n) + 2 Polar nuclei (2n) → Endosperm (3n) → Food reserve for embryo

Pollination Type	Agent	Adaptations / Examples
Autogamy	Self (same flower)	Cleistogamy; pollen shed before flower opens
Anemophily	Wind	Light pollen, feathery stigma — Wheat, Maize, Grass
Entomophily	Insects	Colorful petals, nectar, scent — Rose, Sunflower, Hibiscus
Hydrophily	Water	Reduced perianth, non-sticky pollen — Vallisneria, Hydrilla
Ornithophily	Birds	Red/orange flowers, copious nectar — Bombax, Bottlebrush

1.3 Plant Hormones and Role of Elicitors

Hormone	Site Produced	Primary Effects
Auxin (IAA)	Shoot apex, young leaves	Cell elongation, root initiation, apical dominance, fruit development
Gibberellin (GA)	Young leaves, seeds	Stem elongation, seed germination, bolting, fruit growth

Cytokinin (CK)	Root apex, endosperm	Cell division, shoot differentiation, delays leaf senescence
ABA (Absciscic acid)	Old leaves, fruits	"Stress hormone" — stomata closure, dormancy, abscission
Ethylene (C2H4)	Ripening fruit, aging tissue	Fruit ripening, senescence, abscission of leaves/flowers

Elicitors are signal molecules (biotic — from pathogens; abiotic — UV, heat, wounding) that trigger plant defence: secondary metabolite production (phenols, terpenoids, alkaloids) and Systemic Acquired Resistance (SAR). Used in agriculture to reduce pesticide use.

1.4 & 1.5 Gametophyte, Sporophyte and Alternation of Generations

Plant Group	Dominant Phase	Gametophyte	Sporophyte
Algae	Gametophyte (n)	Free-living, large	Reduced or absent
Bryophytes (Mosses)	Gametophyte (n)	Free-living leafy plant	Dependent on gametophyte
Pteridophytes (Ferns)	Sporophyte (2n)	Small prothallus	The fern plant we see
Gymnosperms	Sporophyte (2n)	Reduced (pollen/embryo sac)	The tree / shrub
Angiosperms	Sporophyte (2n)	Pollen (3-celled) / Embryo sac (7-celled)	The flowering plant

Cycle: Sporophyte (2n) →[Meiosis]→ Spores (n) →[Mitosis]→ Gametophyte (n) →[Gametes + Fertilization]→ Zygote (2n) → New Sporophyte

